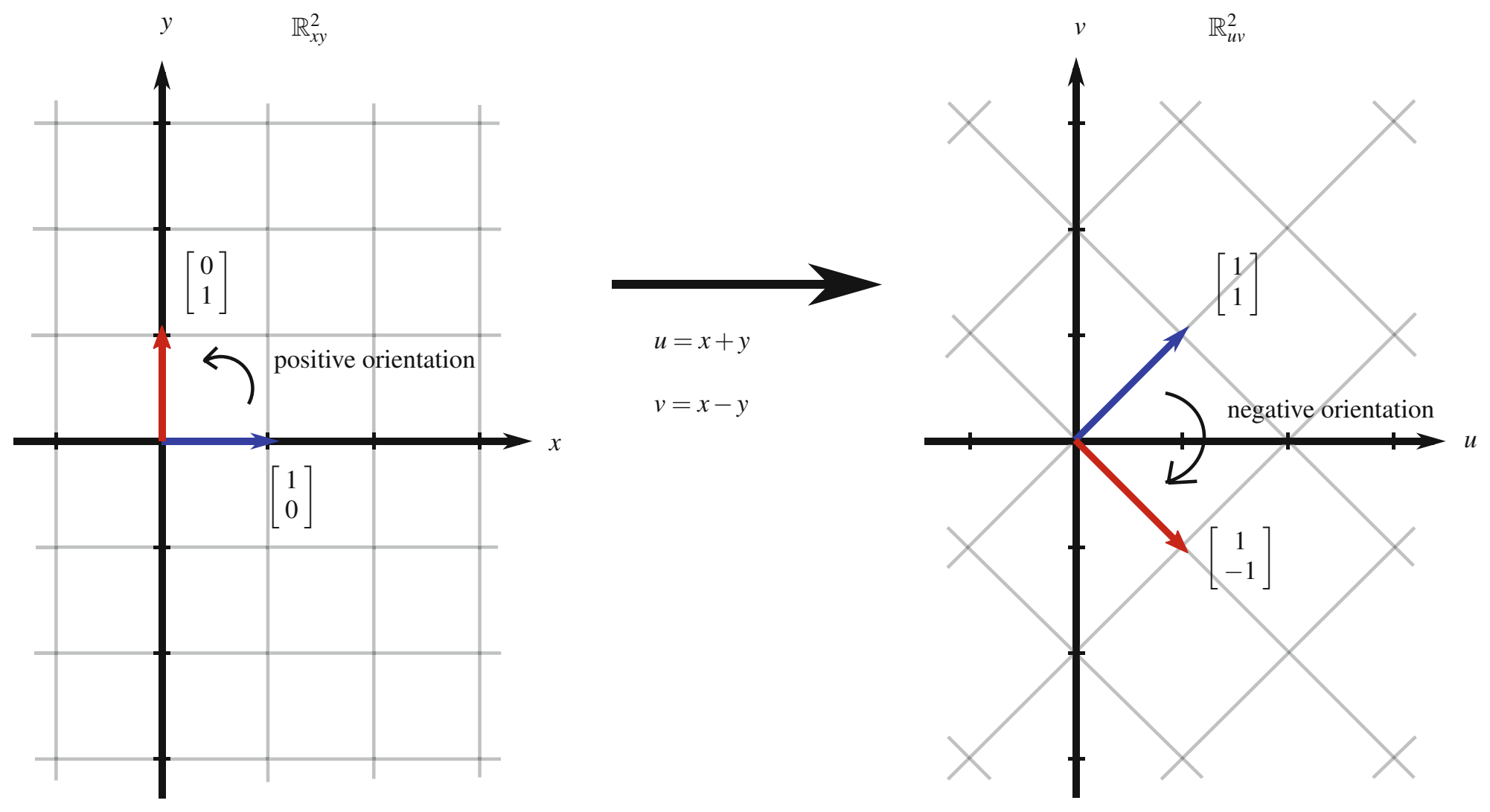

y
$\mathbb{R}^2_{xy}$
$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$
positive orientation
x
$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
$u = x + y$
$v = x - y$
v
$\mathbb{R}^2_{uv}$
$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
negative orientation
u
$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$